CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry 48 (2015) 131–144

Contents lists available at ScienceDirect

CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry

journal homepage: www.elsevier.com/locate/calphad

Thermodynamic reevaluation of the Fe–O system

Taufiq Hidayat [a], Denis Shishin [b], Evgueni Jak [a], Sergei A. Decterov [a,b,*]

[a] Pyrosearch, The University of Queensland, Brisbane, Qld., Australia
[b] Centre de Recherche en Calcul Thermochimique (CRCT), Dép. de Génie Chimique, École Polytechnique, Montréal, Que., Canada

ARTICLE INFO

Article history:
Received 12 September 2014
Received in revised form
9 December 2014
Accepted 22 December 2014
Available online 24 December 2014

Keywords:
Fe–O system
Critical assessment
Wüstite
Magnetite
Metallic and oxide liquid
Hematite

ABSTRACT

The Fe–O system has been assessed over the whole composition range to produce a self-consistent set of thermodynamic properties of all condensed phases from 25 °C to above the liquidus temperatures at ambient pressure. The liquid phase from metallic liquid to oxide melt is described by a single model developed within the framework of the Quasichemical Formalism. The model reflects the existence of strong short-range ordering in oxide liquid at approximately FeO and Fe_2O_3 compositions. Parameters of thermodynamic models have been optimized to reproduce all available thermodynamic and phase equilibrium data. In particular, the heat capacity and entropy of wüstite is described better than in the previous assessments.

1. Introduction

The iron–oxygen system is key to thermodynamic modeling of multicomponent spinel, slag and wüstite, which are important for metallurgy, ceramics and petrology. This system has been the subject of numerous experimental studies and the thermodynamic properties and phase equilibria published up to 1990 are summarized in a comprehensive review by Wriedt [1].

Thermodynamic assessments of the Fe–O system differ mostly in the model for the liquid oxide phase. Goel et al. [2], Bjorkman [3] and Kowalski and Spencer [4] applied the associated solution model; Sundman [5] developed the ionic two-sublattice liquid model, which was later modified [6] to include neutral $FeO_{1.5}$ species on the anion sublattice rather than Fe^{+3} ions on the cation sublattice; and the quasichemical model was used by Wu et al. [7] and Decterov et al. [8].

Two important studies [9,10] of wüstite by adiabatic calorimetry were published after the thermodynamic assessments mentioned above had been completed. The reported low- and high-temperature heat capacities, standard entropies and enthalpies of formation of wüstite as functions of composition substantially constrained the thermodynamic properties of this phase. Since wüstite is in the center of the iron–oxygen system and can be in equilibrium with almost all other phases except for hematite, a complete reevaluation of the whole Fe–O system has to be performed to take into account the new data. This is the subject of the present study.

2. Phases and thermodynamic models

The optimized phase diagram of the Fe–O system is shown in Fig. 1. The major solid solutions are wüstite and spinel, while fcc and bcc iron and hematite, Fe_2O_3, have very narrow ranges of nonstoichiometry. Metallic and oxide liquids are separated by a very large miscibility gap [11–14].

2.1. Wüstite

The literature on wüstite structures is voluminous and only a brief description is included here based on the summary by Wriedt [1]. The structure is a highly defective form of an ideal NaCl-type lattice. Oxygen ions form an fcc lattice and Fe^{2+} cations are located on the interstitial sites. The wüstite solid solution always contains more oxygen than the stoichiometric composition “FeO”. The iron deficiency is normally attributed to formation of neutral vacancies on cation sublattices together with Fe^{3+} cations, which provide charge compensation. Even though the presence of vacancies on the oxygen sublattice cannot be ruled out, the

* Corresponding author at: Centre de Recherche en Calcul Thermochimique (CRCT), Dép. de Génie Chimique, École Polytechnique, Montréal, Que., Canada. Fax: +1 514 340 5840.
E-mail addresses: t.hidayat@uq.edu.au (T. Hidayat), denis.shishin@polymtl.ca (D. Shishin), e.jak@uq.edu.au (E. Jak), sergei.decterov@polymtl.ca (S.A. Decterov).

http://dx.doi.org/10.1016/j.calphad.2014.12.005
0364-5916/

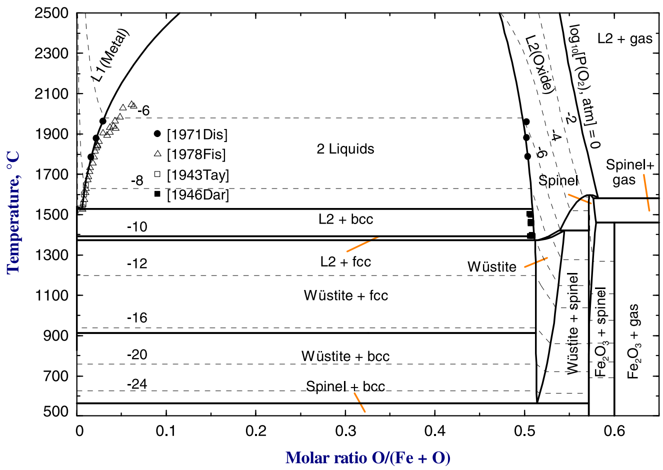

Fig. 1. Optimized phase diagram of the Fe–O system: experimental points [11–14] and calculated lines. Oxygen isobars are shown by the dashed lines.

concentration of these vacancies is definitely much smaller than the ones on the cation sublattice.

Originally it was assumed that iron and vacancies are located only on the octahedral sites, but subsequent detailed neutron diffraction studies indicate that some iron cations, mostly Fe^{3+}, occur interstitially on the tetrahedral sites, which are normally vacant in NaCl-type structures. The basic defect is formed by a tetrahedral Fe ion surrounded by four octahedral vacancies positioned relative to one another at the comers of a tetrahedron. These defects form clusters, in which vacancies are shared among the conjoined defects. Description of the defect structures is further complicated by the incompletely understood long-range ordering of the defect clusters at lower temperatures and higher O/Fe ratios. Three metastable forms of wüstite were observed at and above room temperature, but below the temperature of wüstite decomposition into spinel and bcc iron at about 570 °C. Presumably, these superstructures, which have unit cell dimensions larger than the value of the basic NaCl-type unit cell, originate from long-range ordering of defects.

In the many papers of Vallet and his associates (see, e.g. [15]) it was proposed to subdivide the wüstite field into ten regions separated by transformations of second (or higher) order. These structural changes have not been detected by other authors who worked within the stability field of wüstite and Vallet's findings remain controversial.

Since there is no reliable quantitative information on the occupancies of tetrahedral and octahedral sites and on the compositions and amounts of clusters formed by defects, it is not practical to develop a comprehensive thermodynamic model reflecting the complex structure of wüstite. The most important difference between several simplified models that can be examined is the magnitude of the configurational entropy.

The model proposed by Sundman [5] assumes that iron cations and vacancies mix only on the octahedral sites, ignoring the presence of iron on the tetrahedral sublattice. The formula unit of wüstite is written as $(Fe^{2+}, Fe^{3+}, Va)^{oct}(O^{2-})$. The Compound Energy Model [16] gives the following expression for the Gibbs energy

$$g = (Y_{Fe^{2+}} g^{o}_{Fe^{2+}O^{2-}} + Y_{Fe^{3+}} g^{o}_{Fe^{3+}O^{2-}} + Y_{Va} g^{o}_{VaO^{2-}}) - TS_{config} + g^{ex} \quad (1)$$

$$S_{config} = -R(Y_{Fe^{2+}} \ln Y_{Fe^{2+}} + Y_{Fe^{3+}} \ln Y_{Fe^{3+}} + Y_{Va} \ln Y_{Va}) \quad (2)$$

where $g^{o}_{MO^{2-}}$ is the Gibbs energy of pure end-member MO^{2-}, S_{config} is the configurational entropy, Y_M is the site fraction of species M on the octahedral sublattice and g^{ex} is the excess Gibbs energy which is expanded as a polynomial in the site fractions. This model most likely overestimates the configurational entropy because it neglects formation of the clusters.

Three simple polynomial models can be considered based on the following formula units of the solution:

$$(1-x)FeO + xFeO_{3/2} \quad (3)$$

$$(1-y)FeO + yFe_{2/3}O \quad (4)$$

$$(1-z)FeO + zVaO \quad (5)$$

The first model with the formula unit per mole of cations was used by Fei and Saxena [17] and by Wu et al. [7]. The last two models give thermodynamic properties of wüstite per mole of oxygen.

The Gibbs energy per formula unit of the solution for a polynomial model is given by

$$g = (X_A g^{o}_A + X_B g^{o}_B) - TS_{config} + X_A X_B L_{A,B} \quad (6)$$

$$S_{config} = -R(X_A \ln X_A + X_B \ln X_B) \quad (7)$$

where g^{o}_M and X_M are the Gibbs energy and mole fraction of pure component M, respectively, and S_{config} is the configurational entropy. The molar interaction energy $L_{M,N}$ between components M and N is expanded as a polynomial in the mole fractions of the components:

$$L_{M,N} = \sum_{i,j \geq 0} q^{ij}_{M,N} X^i_M X^j_N \quad (8)$$

where $q^{ij}_{M,N}$ are the binary model parameters.

Model (5) essentially assumes that the configurational entropy originates from the distribution of iron atoms and vacancies on the octahedral sites, that is Fe^{2+} and Fe^{3+} cations are indistinguishable. Even though the changes in configuration due to different arrangements of Fe^{2+} and Fe^{3+} cations occur by electron hopping, which is very fast and does not require diffusion, this model most likely underestimates the configurational entropy. As can be seen from Fig. 2, the configurational entropies of models (3) and (4) are intermediate between those of models (2) and (5). Presumably, the model that provides the best fit to the available experimental data

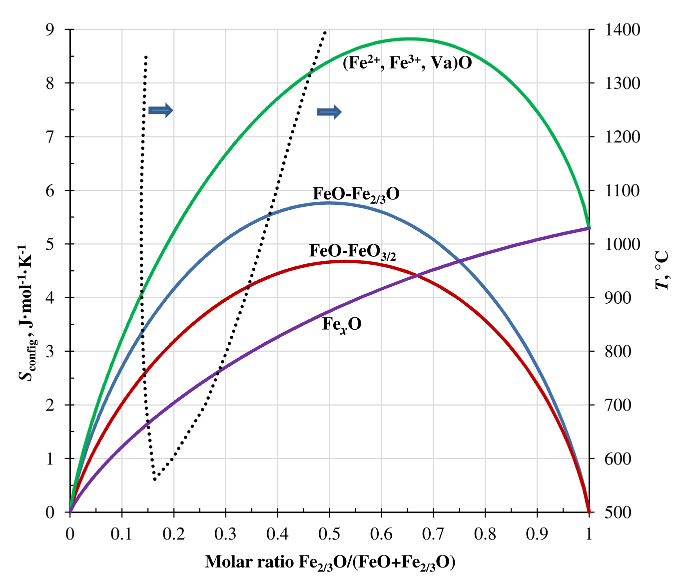

Fig. 2. Configurational entropies of wüstite calculated using different models. The stability range of wüstite at ambient pressure is located between the dotted lines.

has the configurational entropy that is closest to the real one. All four models were tested in the present study.

2.2. Liquid

The Fe–O liquid phase was modeled by Sundman [5] using the ionic two-sublattice model $(Fe^{+2}, Fe^{+3})_P(O^{-2}, Va^{-Q})_Q$ with charged vacancies. Fe^{+3} was later replaced by neutral $FeO_{1.5}$ species [6], resulting in the $(Fe^{+2},)_P(O^{-2}, FeO_{1.5}, Va^{-Q})_Q$ formula unit of the solution. In this form the model was accepted by several authors and used to optimize higher-order systems [18–20].

Kowalski and Spencer [4] suggested a simple associated solution model (Fe, FeO, $FeO_{1.5}$) with one sublattice. Luoma [21] extended their Fe–O solution and presented a thermodynamic database for the Fe–Cr–Ni–C–O system.

Wu et al. [7] and Decterov et al. [8] modeled oxide liquid between the FeO and Fe_2O_3 compositions in the Fe–O system using the modified quasichemical model [22,23]. Metallic melt was modeled as a separate solution [24]. This oxide liquid was further used in many higher-order systems [25–29] as part of the large FactSage *Slag* database [30].

In the present study, the liquid phase was modeled as a single solution over the entire composition range from liquid metal to oxide melt. The thermodynamic model describes drastic changes in the activity of oxygen at the FeO and Fe_2O_3 compositions, which are the result of strong short-range ordering, even though no experimental data are available in the Fe_2O_3 region because of very high equilibrium oxygen pressures. The model for the liquid phase was developed within the framework of the Quasichemical Formalism [22,23]. It has one sublattice containing three species: Fe^{II}, Fe^{III} and O, where Fe^{II} and Fe^{III} correspond to two oxidation states of Fe. In the Fe–O system, the fraction of the Fe^{II}–O pairs goes through a maximum close to the FeO composition, while the Fe^{III}–O pairs are most abundant at the Fe_2O_3 composition. The species are not charged, so the condition of electroneutrality is not imposed and the model is capable to represent the liquid phase from metal to nonmetal.

Figs. 3 and 4 illustrate the concept of two oxidation states of Fe that bring about two compositions of maximum short-range ordering in Fe–O liquid. The calculated enthalpy of mixing between pure liquid iron and hypothetical pure liquid oxygen has two inflections, which correspond to maximum short-range ordering at the FeO and Fe_2O_3 compositions. The special point at the mole fraction of oxygen equal to 0.5 is better seen on the derivative of the enthalpy of mixing. There are no reliable experimental data on the oxidation states of iron in the Fe-O liquid phase, but the drastic transition of Fe^{II} to Fe^{III} between the FeO and Fe_2O_3 compositions seems physically reasonable. The curves in Figs. 3 and 4 were calculated using the optimized model parameters for the liquid phase; for the sake of simplicity, the miscibility gap in the liquid and formation of all other phases were suppressed.

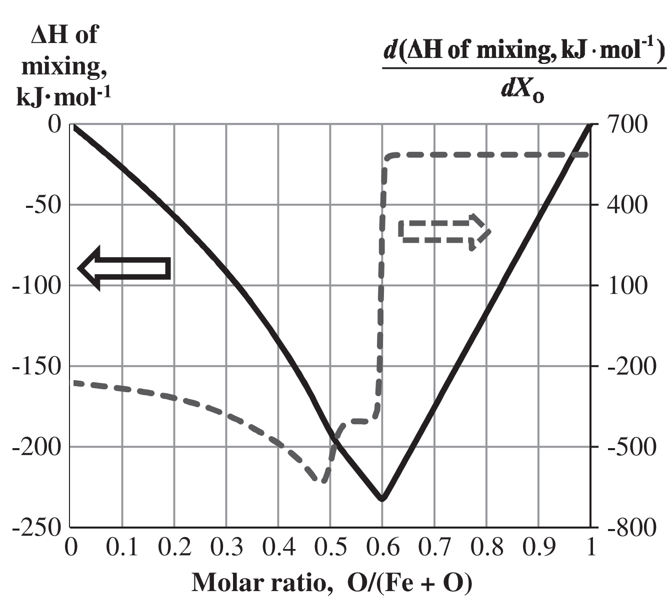

Fig. 3. Calculated enthalpy of mixing (solid line) and its derivative (dashed line) in Fe–O liquid at 1600 °C.

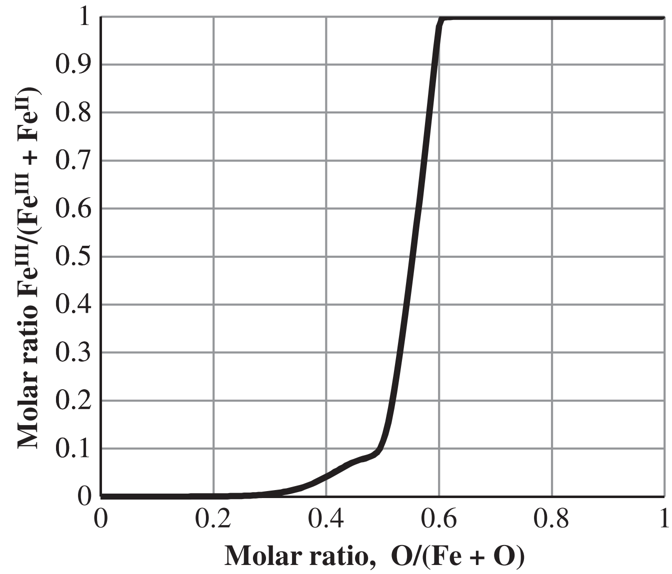

Fig. 4. Calculated distribution of iron between Fe^{II} and Fe^{III} in Fe–O liquid at 1600 °C.

The formulae and notations of the Quasichemical Formalism have been described in detail elsewhere [22,23] and only a brief summary is given here. To explain the meaning of parameters of the Quasichemical Formalism, let us consider a binary solution formed by components A and B. In the pair approximation of the modified quasichemical model, the following pair exchange reaction is considered:

$$(A\text{–}A) + (B\text{–}B) = 2(A\text{–}B);\ \ \Delta g_{AB} \tag{9}$$

where $(i\text{–}j)$ represents a first-nearest-neighbor pair and Δg_{AB} is the non-configurational Gibbs energy change for the formation of two moles of (A–B) pairs.

When the solution is formed from pure components A and B, some (A–A) and (B–B) pairs break to form (A–B) pairs, so the Gibbs energy of mixing is given by [22]:

$$\Delta G_{mix} = G - (n_A g_A^o + n_B g_B^o) = -TS_{config} + \left(\frac{n_{AB}}{2}\right)\Delta g_{AB} \tag{10}$$

where G is the Gibbs energy of the solution, g_A^o and g_B^o are the molar Gibbs energies of pure liquid components; n_A, n_B and n_{AB} are the numbers of moles of A, B and (A–B) pairs and S_{config} is the configurational entropy given by randomly mixing the (A–A), (B–B) and (A–B) pairs. Since no exact expression is known for this entropy of mixing in three dimensions, an approximate equation is used which was shown [22] to be an exact expression for a one dimensional lattice (Ising model) and to correctly reduce to the random mixing point approximation (Bragg Williams model) when Δg_{AB} is equal to zero.

Δg_{AB} can be expanded as an empirical polynomial in terms of the mole fractions of pairs [22]:

$$\Delta g_{AB} = \Delta g^{o}_{AB} + \sum_{(i+j)\geq 1} g^{ij}_{AB} X^{i}_{AA} X^{j}_{BB} \tag{11}$$

where Δg°_{AB} and g^{ij}_{AB} are the parameters of the model which can be functions of temperature. In practice, only the parameters g^{i0}_{AB} and g^{0j}_{AB} need to be included.

The composition of maximum short-range ordering is determined by the ratio of coordination numbers. Let Z_A and Z_B be the coordination numbers of A and B. These coordination numbers can vary with composition as follows [22]:

$$\frac{1}{Z_A} = \frac{1}{Z^{A}_{AA}}\left(\frac{2n_{AA}}{2n_{AA}+n_{AB}}\right) + \frac{1}{Z^{A}_{AB}}\left(\frac{n_{AB}}{2n_{AA}+n_{AB}}\right) \tag{12}$$

$$\frac{1}{Z_B} = \frac{1}{Z^{B}_{BB}}\left(\frac{2n_{BB}}{2n_{BB}+n_{AB}}\right) + \frac{1}{Z^{B}_{BA}}\left(\frac{n_{AB}}{2n_{BB}+n_{AB}}\right) \tag{13}$$

where Z^{A}_{AA} and Z^{A}_{AB} are the values of Z_A when all nearest neighbors of an A are As, and when all nearest neighbors of an A are Bs, respectively, and where Z^{B}_{BB} and Z^{B}_{BA} are defined similarly. For example, in order to set the composition of maximum short-range ordering at the molar ratio $n_A/n_B = 2$, one can set the ratio $Z^{B}_{BA}/Z^{A}_{AB} = 2$. Values of Z^{B}_{BA} and Z^{A}_{AB} are unique to the A–B binary system, while the value of Z^{A}_{AA} is common to all systems containing A as a component.

Even though the model is sensitive to the ratio of the coordination numbers, it is less sensitive to their absolute values. We have found by experience that selecting values of the coordination numbers that are smaller than actual values often yields better results. This is due to the inaccuracy introduced by an approximate equation for the configurational entropy of mixing which is only exact for a one-dimensional lattice. The smaller coordination numbers partially compensate this inaccuracy in the model equations, being more consistent with a one-dimensional lattice. Therefore, the "coordination numbers" in our model are essentially treated as model parameters, which are used mainly to set a physically reasonable composition of maximum short-range ordering.

There are six possible pairs formed by the species Fe^{II}, Fe^{III} and O in the Fe–O liquid phase. The fractions of all pairs are calculated by the Gibbs energy minimization procedure built into the FactSage software [30], using the optimized model parameters. Fe^{II}–Fe^{II} and O–O are the predominant pairs at compositions close to pure Fe and O, respectively. The Fe^{II}–O and Fe^{III}–O pairs are dominant in oxide liquid, whereas the fractions of the Fe^{II}–Fe^{III} and Fe^{III}–Fe^{III} pairs are small at all compositions of interest.

"Coordination-equivalent" fractions (Y_m) are defined as

$$Y_m = Z_m n_m / \sum (Z_i n_i) \tag{14}$$

In addition to the binary terms, the model can have ternary terms, $\Delta g_{AB(C)}$, which give the effect of the presence of component C upon the energy Δg_{AB} of pair exchange reaction (9). Ternary terms are expanded as empirical polynomials either in terms of the mole fractions of pairs (with model parameters $g^{ijk}_{AB(C)}$) or in terms of the "coordination-equivalent" fractions (with model parameters $q^{ijk}_{AB(C)}$). In the Quasichemical Formalism, it is also possible to have Bragg Williams ternary terms, which are essentially the same as $q^{ijk}_{AB(C)}$ terms, except for they are not taken into account in the calculation of the amounts of pairs. Extrapolation of binary terms into the Fe^{II}–Fe^{III}–O system is done by an asymmetric "Toop-like" method [23], taking O as an asymmetric component. The formulae for ternary terms and for extrapolation of binary and ternary terms into a multicomponent system are discussed in detail elsewhere [23].

2.3. Spinel (magnetite)

Magnetite has a cubic spinel structure, space group $Fd\bar{3}m$, prototype $MgAl_2O_4$. The spinel structure can be derived from the fcc close packing of the oxygen ions. Cations occupy half of the octahedral interstices and one-eighth of the tetrahedral interstices. The ideal stoichiometry is Fe_3O_4, where the cations occupy twice as many octahedral sites as tetrahedral ones. The thermodynamic model for spinel was developed within the framework of the Compound Energy Formalism [16] based on the following formula unit of the solution:

$$\left(Fe^{2+}, Fe^{3+}\right)^{tetr}\left[Fe^{2+}, Fe^{3+}, Va\right]^{oct}_{2} O^{2-}_{4} \tag{15}$$

The model is discussed in detail elsewhere [8] and is used without modifications in the present study. The deviation from the ideal stoichiometry of magnetite, Fe_3O_4, toward the metal-rich side is very small. Similar deviations remain small in all multicomponent spinels containing Al, Ca, Co, Cr, Cu, Fe, Mg, Mn, Ni and Zn that have been modeled so far in the FToxid database of FactSage [30]. To the best of our knowledge, such small excess of metals in spinel are not important for any practical applications and were neglected in the present study.

2.4. Solid fcc and bcc iron

Oxygen is soluble to some extent in fcc and bcc iron. The Bragg–Williams random mixing model was used for both phases, which gives the following expression for the molar Gibbs energy:

$$g = (X_{Fe} g^{o}_{Fe} + X_{O} g^{o}_{O}) + RT(X_{Fe}\ln X_{Fe} + X_{O}\ln X_{O}) + X_{Fe} X_{O} L_{Fe,O} \tag{16}$$

where X_i and g^{o}_{i} are the mole fraction and molar Gibbs energy of component i, $L_{i,j}$ represents the interaction energy between i and j, which can be a function of temperature and composition.

2.5. Fe_2O_3 (hematite)

Hematite has a trigonal structure based on the hcp oxygen-packing scheme. The space group is $R\bar{3}c$ [31]. Hematite is slightly nonstoichiometric (oxygen-deficient) at high temperatures in equilibrium with magnetite, but the exact deviations from the ideal Fe_2O_3 stoichiometry reported by different authors are contradictory and less than 1 mol% [1]. This nonstoichiometry was neglected in the present study. Hematite is antiferromagnetic below a Neel temperature of 956 K.

3. Critical evaluation of experimental data and optimization

The major reason for the present reevaluation of the Fe–O system is to take into account the new experimental data for wüstite [9,10] and to introduce a new model for the liquid phase that is based on the Quasichemical Formalism and describes both metal and oxide liquids.

The properties of spinel and hematite were optimized earlier [8] and only small adjustments were required to make them completely consistent with the new wüstite and liquid.

Most of the experimental data on the Fe–O system are summarized in the figures presented in this section and compared to the calculated lines that are based on the optimized model parameters. Some additional data on spinel and hematite can be found in the earlier optimization [8]. A few experimental studies that are

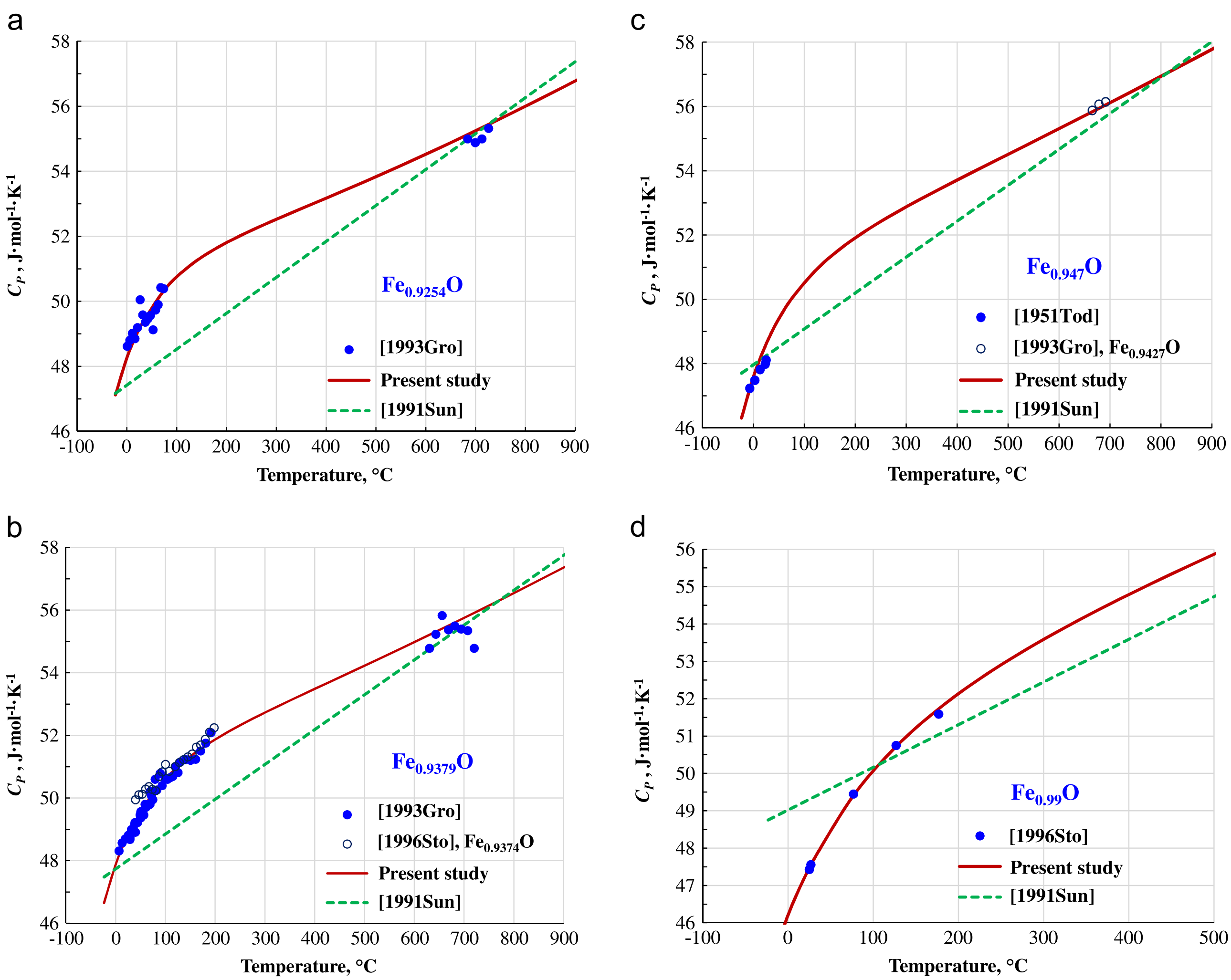

Fig. 5. Heat capacity of wüstite at different compositions: experimental points [9,10,32], the assessment of Sundman [5] and calculated lines.

either inconsistent with the other authors or do not provide substantially new information compared to more accurate measurements are not shown in the figures. References to these studies can be found in reviews by Wriedt [1] and by Kowalski and Spencer [4]. The optimization of Sundman and Selleby [5,6] is the most comprehensive optimization of the Fe–O system presently available. It is compared with the results of the present optimization wherever the difference between the two optimizations is substantial.

3.1. Wüstite

3.1.1. Heat capacity

The low-temperature heat capacity of wüstite was measured by adiabatic calorimetry for several compositions shown in Fig. 5 [9,32]. A different adiabatic calorimeter was used to measure the heat capacity in a narrow temperature range around 700 °C [9]. The heat capacity of wüstite of the $Fe_{0.99}O$ composition was derived from the adiabatic calorimetry measurements on three-phase samples containing Fe and Fe_3O_4 apart from wüstite [10]. In addition, Coughlin et al. [33] measured the heat content from 25 to 1511 °C for the same sample that was used by Todd and Bonnickson [32] for the low-temperature experiments. These data are shown in Fig. 6. The heat content measurements on wüstite

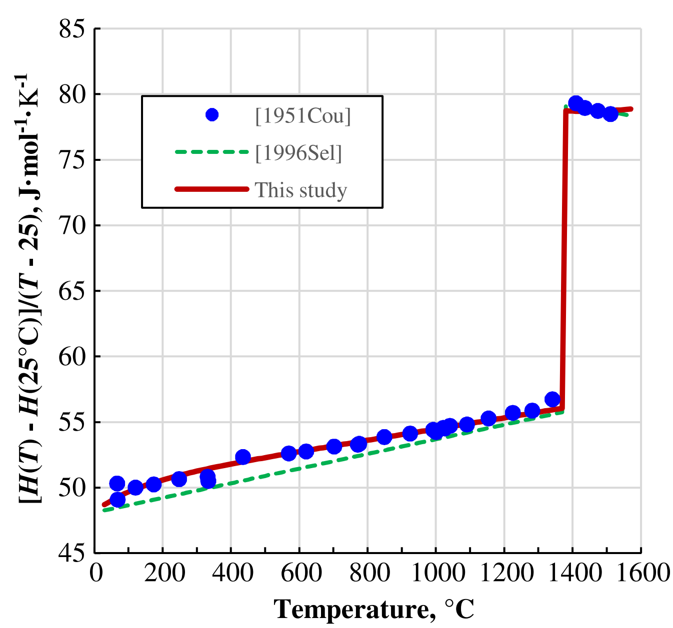

Fig. 6. Reduced enthalpy increment of wüstite $Fe_{0.947}O$: experimental points [33], the assessment of Sundman [5] and calculated lines.

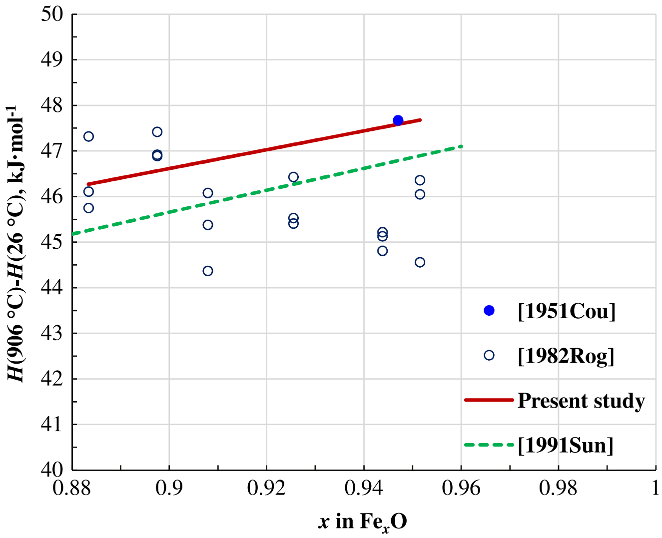

Fig. 7. Heat contents of wüstite at different compositions: experimental points [33,34], the assessment of Sundman [5] and calculated lines.

samples of several compositions were also reported by Rogez et al. [34]. As can be seen from Fig. 7, these data are scattered and tend to be lower than the results of Coughlin et al. [33] at higher concentrations of Fe in wüstite.

In the present study, the heat capacities of the FeO and $Fe_{2/3}O$ end-members of the wüstite solution were fitted to describe all available data on the heat capacities and heat contents of wüstite according to the following equation, which gives C_P per one mole of oxygen:

$$C_P(Fe_xO) = C_P\left(Fe_{1-y/3}O\right) = (1-y)C_P(FeO) + yC_P\left(Fe_{2/3}O\right) \qquad (17)$$

As can be seen from Figs. 5 to 7, all the data are well reproduced without using any excess heat capacity terms.

3.1.2. Equilibrium oxygen partial pressure

The numerous measurements of the equilibrium oxygen partial pressure over the wüstite stability field [35–46] are generally in good agreement. As can be seen from Fig. 8, the experimental points are well reproduced by the wüstite model optimized in the present study. The measurements below about 800 °C are less accurate due to slow diffusion and equilibration.

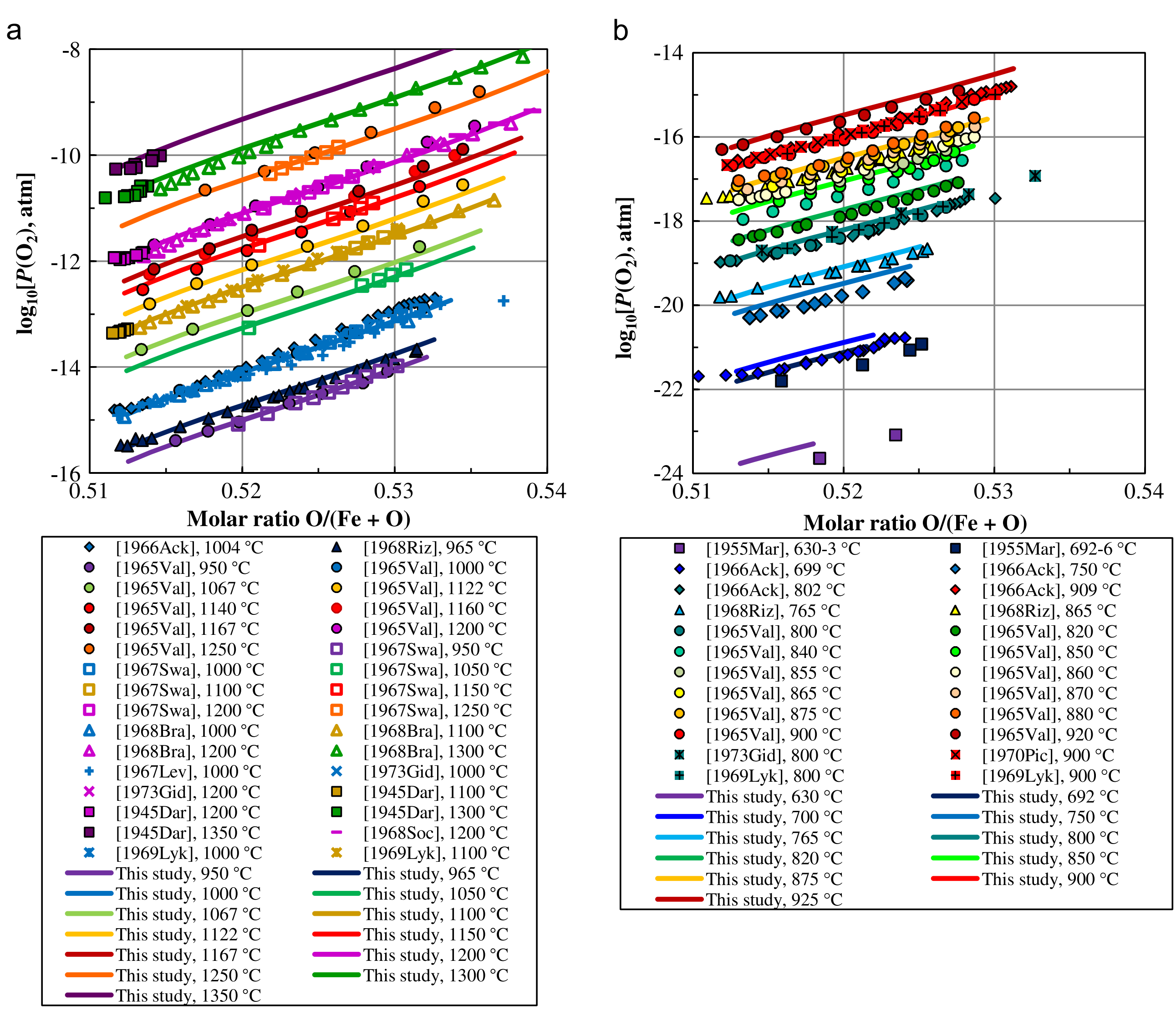

Fig. 8. Equilibrium oxygen partial pressures over the wüstite stability field: experimental points [35–46] and calculated lines.

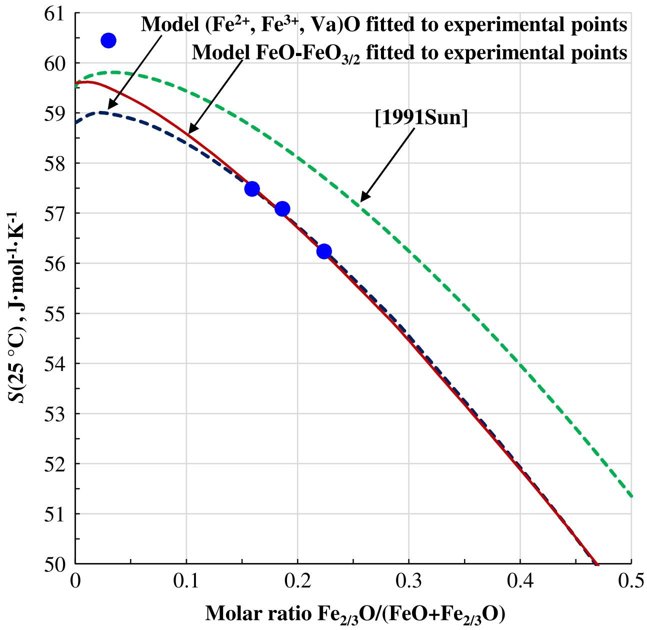

Fig. 9. Standard entropy at 25 °C: experimental points [9,10,32], the assessment of Sundman [5] and calculated lines. The solid line represents the model that was accepted in the present study.

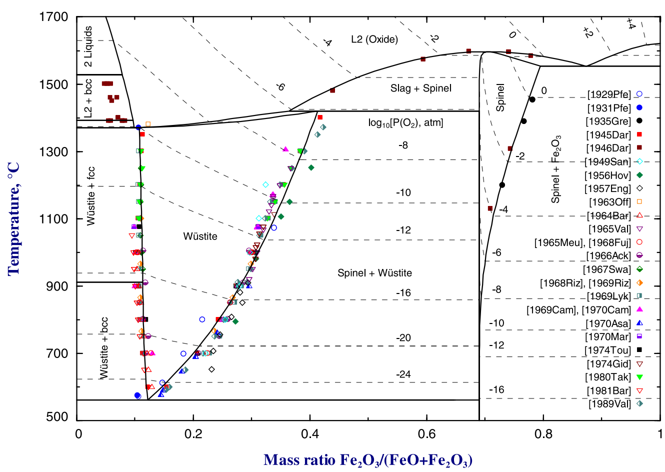

Fig. 10. Optimized phase diagram of the Fe–O system in the $FeO–Fe_2O_3$ region: experimental points [15,36,37,41,42,45,52–67] and calculated lines. The calculated oxygen isobars are shown by the dashed lines.

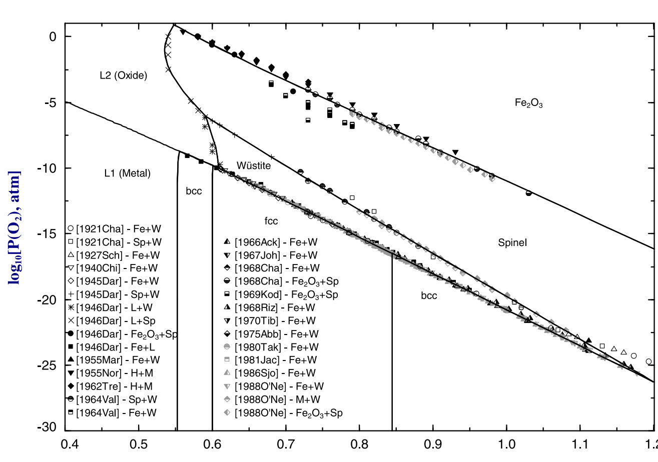

Fig. 11. Potential phase diagram of the Fe–O system: experimental points [35–37,45,66,68–80] and calculated lines.

3.1.3. Entropy and enthalpy

Standard entropies of metastable wüstite at four different compositions were obtained by integrating from 0 K to room temperature the low-temperature heat capacities measured for single-phase samples [9,32] and a three-phase sample containing wüstite of the $Fe_{0.99}O$ composition [10]. The value derived for $Fe_{0.99}O$ wüstite is clearly much less accurate due to uncertainties in the amounts of phases present in the sample and the accumulation of errors resulted from the subtraction of the entropies of pure Fe and Fe_3O_4 from the total entropy of the sample. The reported entropies are shown in Fig. 9.

Todd and Bonnickson [32] argued that the entropy of non-stoichiometric wüstite samples at 0 K should not be zero but rather equal to the residual configurational entropy. The results of the present study demonstrated that the entropies obtained by integration of the low-temperature heat capacities are in better agreement with the high-temperature data if S (0 K) is close to zero.

The thermodynamic properties of wüstite over its stability range are well constrained by the experimental data. In particular, the numerous studies of equilibrium oxygen partial pressure over wüstite shown in Fig. 8 in combination with the accurate data for $P(O_2)$ in the wüstite+iron region discussed in the next section allow obtaining the chemical potential of iron in wüstite by model-independent integration of the Gibbs–Dughem equation and hence calculating the Gibbs energy of wüstite as a function of temperature and composition. Therefore, the entropy at room temperature can be calculated from the high-temperature data and the experimental heat capacities of wüstite discussed in Section 3.1.1. This resulted in the values that are consistent with the assumption that S (0 K)=0 in the low-temperature heat capacity experiments. Presumably, some ordering eliminating the configurational entropy is superimposed over the magnetic order-disorder transition around the Néel temperature, which is about −80 °C.

If the data shown in Fig. 9 are used to calibrate a thermodynamic model for wüstite, all the models discussed in Section 2.1 give almost identical results. Whether a model assumes the higher or lower configurational entropy, e.g. Eq. (2) or Eq. (7), the lattice entropies of pure end-members of the solution have to be adjusted accordingly for the calculated curve to pass through the experimental points, so that the total calculated entropy becomes almost identical over the stability range of wüstite. The models with low configurational entropy describe a little better the experimental value for the $Fe_{0.99}O$ composition, which is outside the stability range of wüstite. In the present study, the polynomial model with formula unit (3) of the solution was accepted.

The three wüstite samples of different compositions that were used for measuring heat capacities were decomposed to iron and magnetite at about 525 °C and then recombined again in the calorimeter [9]. The formation of wüstite started at about 577 °C and required increased temperature and several days for completion. The enthalpies of formation of wüstite were evaluated from these results. The partial enthalpies of oxygen were measured in a Tian-Calvet-type microcalorimeter at 800 °C [47] and 1075 °C [48]. Both the integral and partial enthalpies are relatively less accurate than the equilibrium oxygen partial pressures discussed in the previous section due to experimental difficulties. The enthalpy data [9,47,48] are reproduced by the wüstite model optimized in the present study within the experimental error limits.

3.1.4. Phase diagrams

The optimized phase diagram of the Fe–O system is shown in Fig. 1 and the $FeO–Fe_2O_3$ region of this diagram can be seen in detail in Fig. 10.

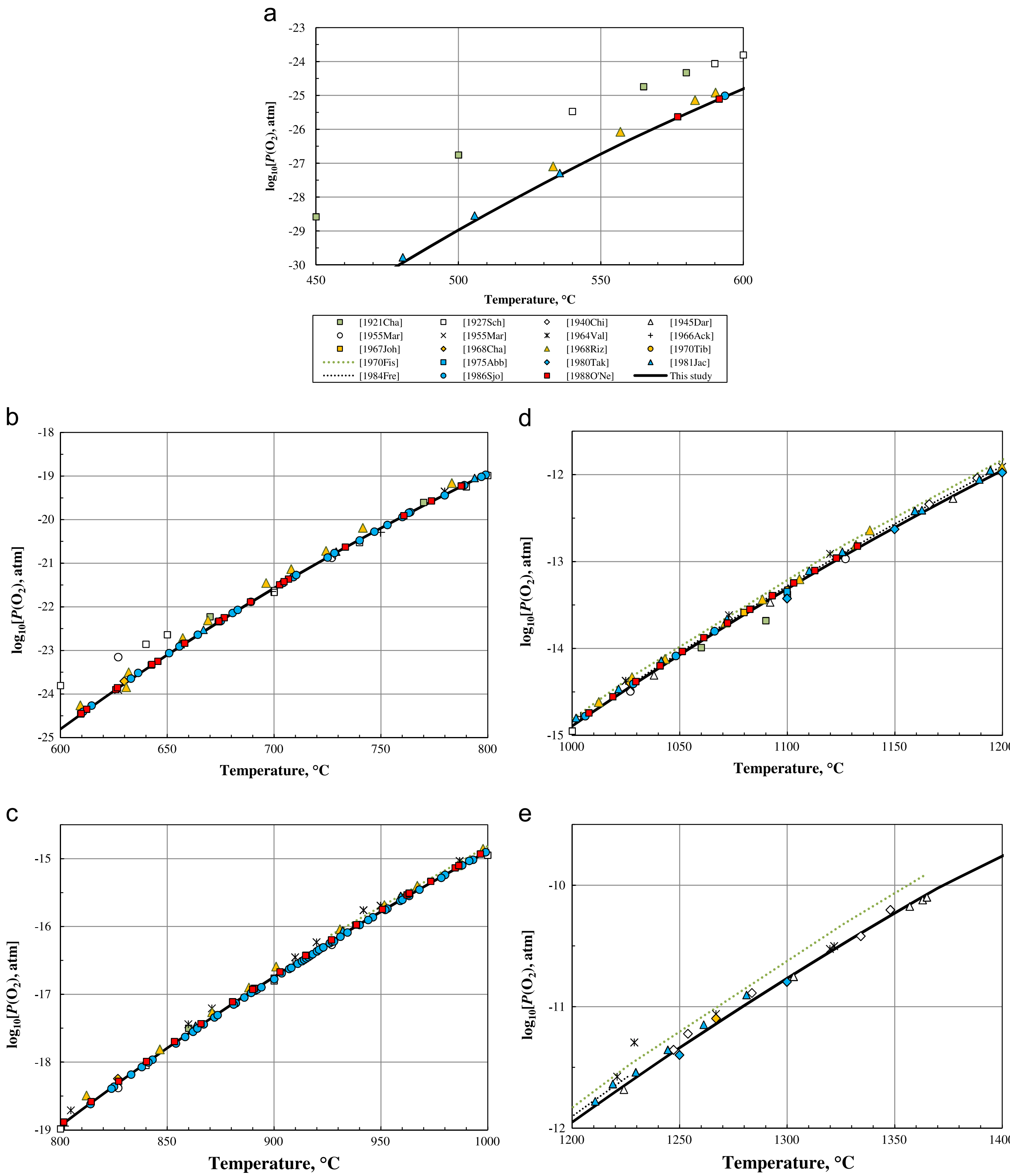

Fig. 12. Oxygen potential in the iron (bcc or fcc)+wüstite two-phase fields: experimental points [35–37,45,66,68–70,72–74,76–82] and calculated line.

The phase boundaries of the wüstite stability field were reported in numerous studies, which were reviewed by Wriedt [1]. The data that were considered by Wriedt to be the most accurate are shown in Fig. 10.

The calculated phase boundary of oxide liquid in equilibrium with solid iron is slightly shifted to the oxygen-rich side from the experimental points reported by Darken and Gurry [11]. However, this phase boundary was measured in the Fe–O–S system in three different studies [49–51], which are in good agreement. Extrapolation of these results to the Fe–O system suggests the location of the Fe+L2 boundary at higher oxygen concentrations than reported by Darken and Gurry [11]. Even though it was possible to reproduce these points in Fig. 10 quite well by adjusting the model parameters for the liquid phase, this adversely affected the description of the Fe–O–S system. Considering the scatter of the data of Darken and Gurry [11], the preference was given to the three

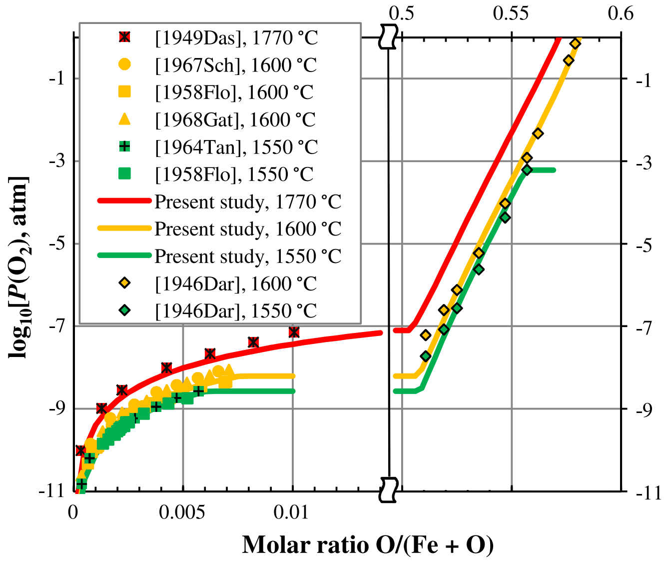

Fig. 13. Oxygen potential in Fe–O liquid expressed as isotherms: experimental points and calculated lines [11,83–88].

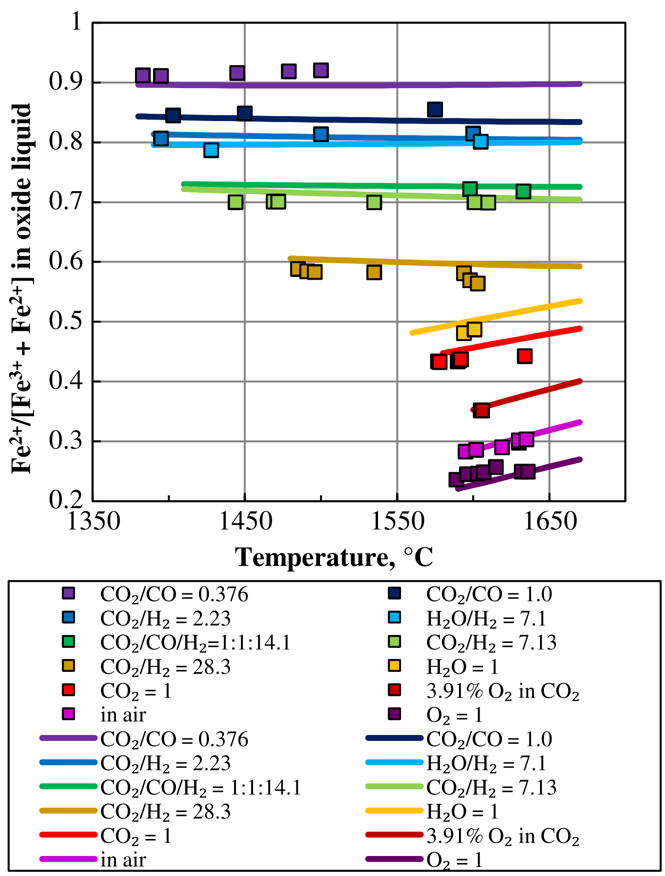

Fig. 14. Compositions of oxide liquid equilibrated with various gas phases: experimental points [11] and calculated lines.

studies in the Fe–O–S system.

The potential phase diagram of the Fe–O system is shown in Fig. 11. The oxygen potential in the iron (bcc or fcc)+wüstite two-phase fields has been studied most thoroughly. These data are of particular importance because Fe+wüstite mixtures are often used as a reference electrode in EMF measurements. The expanded scale of Fig. 12 makes it possible to see the experimental scatter of

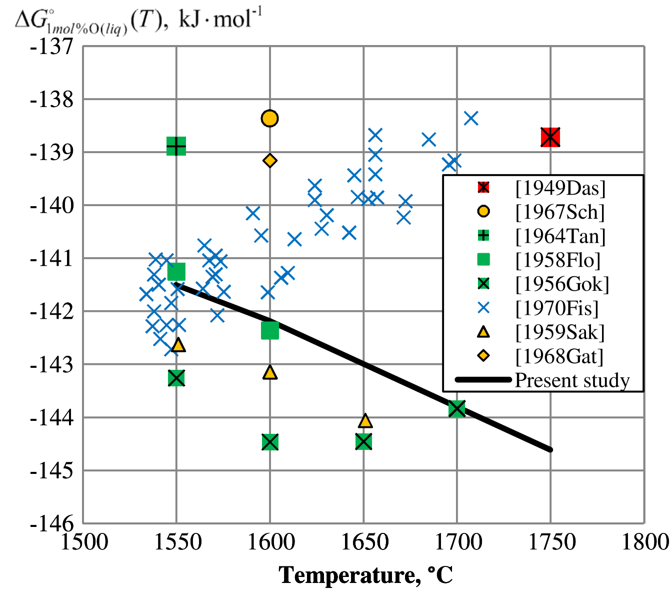

Fig. 15. Gibbs energy of dissolution of oxygen in liquid Fe at infinite dilution with 1 mol% O liquid and 1 atm O_2 gas as reference states: experimental points [83–90] and calculated line.

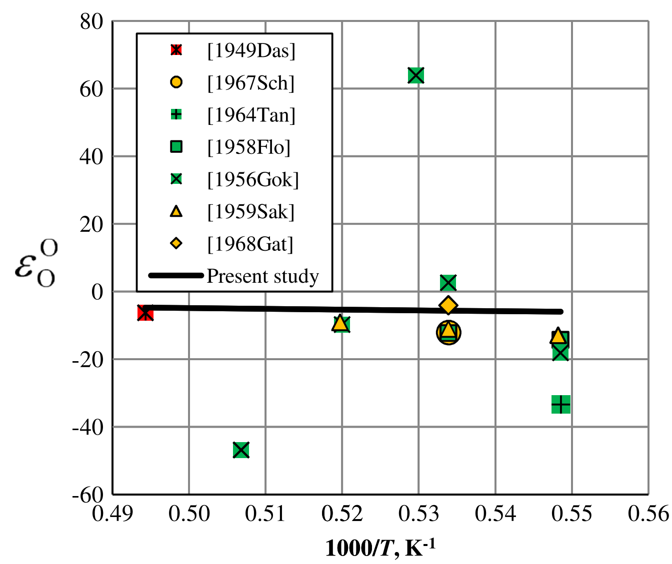

Fig. 16. First-order interaction coefficient for O in liquid Fe: experimental points [83–89] and calculated line.

these data and to compare the data points with the calculated lines.

3.2. Liquid phase

A drastic increase of the oxygen potential is observed in the liquid phase between the FeO and Fe_2O_3 compositions, as can be seen from Fig. 13. Sundman [5] assessed the data [83–86] on the activity of oxygen in liquid iron up to $X_O = 0.015$. Additional experimental data of Floridis and Chipman [87], Goksen [88], Sakao and Sano [89] and Fischer et al. [90] were considered in the present study. The experimental results are expressed as isotherms $\log_{10} P(O_2)$ vs X_O and shown in Fig. 13.

The equilibrium measurements of Darken and Gurry [11] for oxide liquid were performed at $P(O_2)$ fixed by passing a mixture of two gases (CO/CO_2, CO_2/H_2, H_2/H_2O, O_2/N_2) over the condensed

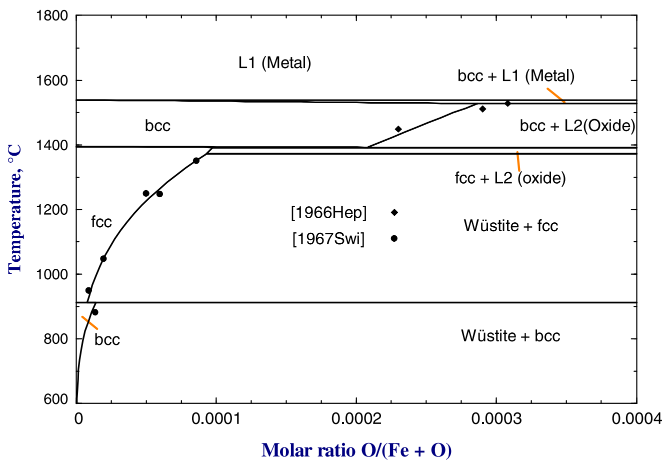

Fig. 17. Solubility of oxygen in fcc and bcc (solid Fe) phases: experimental points [93,94] and calculated lines.

phase at constant temperature. Their original data are shown in Fig. 14. These data were also interpolated in the present study to $\log_{10} P(O_2)$ vs X_O isotherms and plotted in Fig. 13.

It is difficult to evaluate from Fig. 13 the real scatter of the experimental data in the metallic region. To compare the measured solubilities of oxygen in liquid iron at different temperatures and compositions, these experimental data are normally presented as follows. The equality of the chemical potentials of oxygen in liquid iron and in the equilibrium gas phase can be written as

$$\mu_O = 0.5\mu^{+}_{O_2}(T) + RT\ \ln(\gamma_O X_O) = 0.5[\mu^{+}_{O_2}(T) + RT\ \ln P(O_2)] \tag{18}$$

where $\mu^{+}_{O_2}$ is the Gibbs energy of ideal O_2 gas at 1 atm, γ_O is the activity coefficient of oxygen in the liquid phase and $P(O_2)$ is the equilibrium oxygen partial pressure in atm. Hence

$$\ln \gamma_O = 0.5 \ln P(O_2) - \ln X_O \tag{19}$$

In the Fe-rich region up to X_O about 0.01, $\ln \gamma_O$ changes almost linearly with composition. It is a common practice to recalculate the experimental data in this region into $\ln \gamma_O$ and express it as a linear function:

$$\ln \gamma_O = \ln\left(\gamma^o_O(T)\right) + \varepsilon^O_O(T) X_O \tag{20}$$

where $\gamma^o_O(T)$ is the activity coefficient of oxygen at infinite dilution and $\varepsilon^O_O(T)$ is the first-order interaction coefficient:

$$\gamma^o_O(T) = \lim_{X_O \to 0}\left[\frac{P(O_2)^{1/2}}{X_O}\right] \tag{21}$$

In turn, $\gamma^o_O(T)$ is often recalculated into the "Gibbs energy of dissolution of oxygen in liquid iron at infinite dilution with 1 mol% O liquid and 1 atm O_2 gas as reference states" defined as:

$$\Delta G^o_{1\ \mathrm{mol\%}O\ (\mathrm{liq})}(T) \equiv RT \ln\left(\frac{\gamma^o_O(T)}{100}\right) \tag{22}$$

Substitution of Eq. (21) into Eq. (22) gives

$$\Delta G^o_{1\ \mathrm{mol\%}O\ (\mathrm{liq})}(T) = \lim_{X_O \to 0}\left[RT \ln\left(\frac{\sqrt{P(O_2),\ \mathrm{atm}}}{100 X_O}\right)\right] \tag{23}$$

$\Delta G^o_{1\ \mathrm{mol\%}O\ (\mathrm{liq})}(T)$ is a convenient form of presenting experimental results because it is a nearly linear function of temperature. The first-order interaction coefficient, $\varepsilon^O_O(T)$, is usually close to a linear function of inverse temperature (in K). These functions are plotted in Figs. 15 and 16. The lines calculated from the model parameters optimized in the present study are shown along with the selected experimental data [83–90]. The scatter of the experimental points is clearly visible in Figs. 15 and 16.

The data points of Fischer et al. [90] in Fig. 15 suggest the slope of the temperature dependence that could not be reconciled with the miscibility gap data shown in Fig. 1. The properties of oxide liquid are well constrained by the phase diagram data (Figs. 10 and 11) in combination with the thermodynamic properties of wüstite and the equilibrium oxygen partial pressures over the liquid phase shown in Fig. 14. If the results of Fischer et al. [90] are accepted, the miscibility gap on the iron side becomes too wide, deviating from the experimental points shown in Fig. 1 towards pure Fe.

The results of the present optimization are almost identical to the assessment of the solubility of oxygen in liquid iron reported in the Steelmaking Data Sourcebook [91]. In the technical literature, Eqs. (18)–(23) are normally written using weight rather than mole fractions [91]:

$$\mu_O = 0.5\mu^{+}_{O_2}(T) + RT\ \ln(f_O W_O) = 0.5[\mu^{+}_{O_2}(T) + RT\ \ln P(O_2)] \tag{24}$$

$$\log_{10}(f_O) = \log_{10}\left(f^o_O(T)\right) + e^O_O(T)(W_O \times 100) \tag{25}$$

$$\Delta G^o_{1\ \mathrm{wt\%}O\ (\mathrm{liq})}(T) \equiv RT \ln\left(\frac{f^o_O(T)}{100}\right) = \Delta G^o_{1\ \mathrm{mol\%}O\ (\mathrm{liq})}(T) + RT \ln \frac{M_{Fe}}{M_O} \tag{26}$$

$$e^O_O(T) = \frac{\log_{10} e}{100}\left[(\varepsilon^O_O - 1)\frac{M_{Fe}}{M_O} + 1\right] \tag{27}$$

where W_O is the weight fraction of oxygen, M_i is the atomic mass, $\Delta G^o_{1\ \mathrm{wt\%}O\ (\mathrm{liq})}$ is the "Gibbs energy of dissolution of oxygen in liquid iron at infinite dilution with 1 wt% O liquid and 1 atm O_2 gas as reference states", f_O is the activity coefficient of oxygen and e^O_O is the first-order interaction coefficient in weight notations. The values of $\Delta G^o_{1\ \mathrm{wt\%}O\ (\mathrm{liq})}(T)$ and $e^O_O(T)$ tabulated in the Steelmaking Data Sourcebook [91] were recalculated into $\Delta G^o_{1\ \mathrm{mol\%}O\ (\mathrm{liq})}(T)$ and $\varepsilon^O_O(T)$ for comparison with the present study.

3.3. Solid fcc and bcc iron

The solubility of oxygen in solid Fe, which has the fcc or bcc structure, is shown in Fig. 17. The experimental data were assessed by Kowalski and Spencer [4]. One temperature-dependent model parameter, $L_{Fe,O}$, was optimized for each phase to reproduce these data. The choice of the Gibbs energy of hypothetical oxygen with the fcc structure, g^o_O, was explained in the previous article [92]. The same function was used for hypothetical oxygen with the bcc structure. In principle, these Gibbs energies should be somewhat different, but since the solubility of oxygen in iron is very small, this difference is not important for all practical purposes because it is compensated by the value of the $L_{Fe,O}$ parameter that is used to fit the experimental data.

3.4. Optimization

The optimized parameters of the thermodynamic models for the phases in the Fe–O system are given in Table 1. The techniques of thermodynamic evaluation, optimization and modeling are discussed in more detail elsewhere [97].

First, the heat capacity, heat content and standard entropy data for wüstite shown in Figs. 5–7 and 9 were fitted by optimizing the heat capacities and entropies of pure FeO and Fe_2O_3 end-members of the solution. No excess heat capacity or entropy terms were used. Then the enthalpies of formation of the wüstite end-members, FeO and Fe_2O_3, and two temperature-independent interaction parameters were optimized to describe the equilibrium

Table 1
Optimized model parameters for the liquid and solid phases in the Fe–O system.

Phase	Temperature range (K) and reference	Molar Gibbs energy $g(T)$, J mol^{-1}
Wüstite ($Fe_{1-x}O$)	Bragg–Williams (FeO, $FeO_{1.5}$)	
g^{o}_{FeO}	298 – 2500	$-285203.5+274.2455\,T-49.19444T\ln T-0.004678477\,T^{2}+297568.8\,T^{-1}+574.4469\ln T$
$g^{o}_{FeO_{1.5}}$	298 – 2500	$-523138.0+73.37019\,T-26.96809T\ln T-0.008835071\,T^{2}+1498519\,T^{-1}+25471.09\ln T$
$q^{00}_{FeO,\,FeO_{1.5}}$		-59412.8
$q^{10}_{FeO,\,FeO_{1.5}}$		42676.8
Liquid (from metal to oxide)	Modified Quasichemical Model: (Fe^{II}, Fe^{III}, O); Grouping: Fe^{II}, Fe^{III} in group 1 and O in group 2	
$Z^{Fe^{II}}_{Fe^{II}Fe^{II}}=Z^{Fe^{III}}_{Fe^{III}Fe^{III}}=Z^{O}_{OO}=Z^{Fe^{II}}_{Fe^{II}Fe^{III}}=Z^{Fe^{III}}_{Fe^{III}Fe^{II}}=6,\ Z^{Fe^{II}}_{Fe^{II}O}=2,\ Z^{o}_{OFe^{II}}=2,\ Z^{o}_{OFe^{III}}=3,\ Z^{Fe^{III}}_{Fe^{III}O}=2$		
g^{o}_{O}	298–2990 [92]	$121184.8+136.0406\,T-24.50000T\ln T-9.8420\times10^{-4}T^{2}-0.12938\times10^{-6}T^{3}+322517\,T^{-1}$
$g^{o}_{Fe^{II}}$	298 – 1811 [95]	$13265.9+117.5756\,T-23.5143T\ln T-0.00439752\,T^{2}-5.89269\times10^{-8}T^{3}+77358.5\,T^{-1}-3.6751551\times10^{-21}T^{7}$
	1811–6000 [95]	$-10838.8+291.3020\,T-46.0000T\ln T$
$g^{o}_{Fe^{III}}$	298 – 6000	$g^{o}_{Fe^{II}}+6276.0$
$\Delta g^{o}_{Fe^{II}O}$		-391580.56
$g^{10}_{Fe^{II}O}$		$129778.63-30.3340T$
$\Delta g^{o}_{Fe^{III}O}$		$-394551.2+12.5520T$
$g^{20}_{Fe^{III}O}$		83680.00
$q^{001}_{Fe^{II}Fe^{III}(O)}$	Bragg–Williams parameter	$30543.20-44.0041T$
$\Delta g^{o}_{Fe^{II}Fe^{III}}$		83680.00
fcc (Solid Fe)	Bragg-Williams (Fe, O)	
g^{o}_{Fe}	298–1811 [95]	$-236.5+132.4156\,T-24.6643T\ln T-0.003758\,T^{2}+77359.0\,T^{-1}-5.8927\times10^{-8}T^{3}$
	1811–6000 [95]	$-27097.2+300.2521\,T-46.0000T\ln T-2.78854\times10^{31}T^{-9}$
g^{o}_{O}	298–2000 [92]	$120184.8+139.1406T-24.5000T\ln T-9.8420\times10^{-4}T^{2}-0.12938\times10^{-6}T^{3}+322517T^{-1}$
$L_{Fe,O}$		$-315652.63336+27.6144T$
Magnetic properties of Fe	[96]	$T_{Neel}=67$ K Magnetic moment $\beta=0.70$ Structure-dependent parameter $P=0.28$
bcc (Solid Fe)	Bragg–Williams (Fe, O)	
g^{o}_{Fe}	298–1811 [95]	$10375.2+114.5502\,T-23.5143T\ln T-0.004398\,T^{2}+77359.0\,T^{-1}-5.8927\times10^{-8}T^{3}$
g^{o}_{O}	Same as in fcc [92]	
$L_{Fe,O}$		$-315149.19+20.6935T$
Magnetic properties of Fe	[96]	$T_{Curie}=1043$ K Magnetic moment $\beta=2.22$ Structure-dependent parameter $P=0.40$
Magnetite (Spinel, $Fe_{3-x}O_4$)	Compound Energy Model (Fe^{2+}, Fe^{3+}) [Fe^{2+}, Fe^{3+}, Va]$_2$O$_4$, all other model parameters for spinel are reported in Reference [8]	
$g^{o}_{Fe_3O_4}$	298 – 2500	$-1140237+1015.067\,T-0.008149196T^{2}-174.832T\ln T+1445276T^{-1}$
Hematite (Fe_2O_3)	Stoichiometric compound	

Table 1 (*continued*)

fcc (Solid Fe)	Bragg-Williams (Fe, O)	
$g^{o}_{Fe_2O_3}$	298 − 2500	$-859683.1+828.0501T-137.0089T\ln T+1453820T^{-1}$
	2500 − 4000	$-857356.9+823.7122T-136.5437T\ln T$
Magnetic properties of Fe_2O_3		$T_{Neel}=955.667$ K Magnetic moment $\beta=8.36667$ Structure-dependent parameter $P=0.28$

The thermodynamic properties of gaseous species were taken from the FactSage pure substance database [30].

oxygen partial pressures over the wüstite stability field (Fig. 8), the wüstite phase boundaries (Fig. 10) and the oxygen potentials in the (bcc + wüstite) and (fcc + wüstite) fields (Fig. 12).

The thermodynamic properties of phases are optimized in the present study from the room temperature up to about 2000 °C. If the properties of wüstite are extrapolated slightly below the room temperature, the model predicts a miscibility gap with a critical point at the $Fe_{0.955}O$ composition at about 12 °C. This is to be expected for a phase with slightly positive interactions, which is common for solid solutions. The optimization of Sundman [5] predicts the miscibility gap in wüstite with a critical point at the $Fe_{0.94}O$ composition and 274 °C. It should be noted that metastable samples of wüstite quenched from high temperatures into the miscibility gap region cannot decompose into two phases with different compositions due to very slow diffusion at these temperatures, so it is essentially impossible to observe this miscibility gap experimentally. Clearly, any bulk thermodynamic properties measured on quenched samples of wüstite, such as low-temperature heat capacities, cannot be affected by the existence of the miscibility gap.

The thermodynamic properties of magnetite, $Fe_{3-x}O_4$, and hematite, Fe_2O_3, were slightly changed as compared to the earlier optimization [8] to reproduce the wüstite-spinel and spinel-Fe_2O_3 boundaries in the potential phase diagram (Fig. 11) and the spinel phase boundary with hematite in Fig. 10. The enthalpy of formation and standard entropy of the Fe_3O_4 end-member of spinel and the enthalpy of formation of hematite were adjusted within the corresponding experimental error limits. This did not cause any noticeable changes in the description of all other data on spinel and hematite that are discussed in the previous optimization [8].

Each of bcc and fcc iron solid solutions required one optimized temperature-dependent interaction parameter to describe the oxygen solubility limits shown in Fig. 17.

Finally, the thermodynamic properties of the liquid phase were optimized to reproduce the phase diagrams in Figs. 1, 10 and 11, the equilibrium oxygen potentials in Figs. 13 and 14, and the solubility of oxygen in liquid iron (Figs. 15 and 16).

The end members of the liquid phase are pure liquid Fe^{II}, Fe^{III} and hypothetical pure liquid atomic oxygen, which should not be confused with real molecular liquid oxygen. The Gibbs energy of pure liquid iron, $g^{o}_{Fe^{II}}$, was taken from the SGTE database [95]. $g^{o}_{Fe^{III}}$ is essentially the same, but a positive value was added to suppress formation of Fe^{III} in pure Fe (see Fig. 4).

The Gibbs energy of pure liquid oxygen, g^{o}_{O}, was estimated in the previous study [92]. Large negative parameters $\Delta g^{o}_{Fe^{II}O}$ and $\Delta g^{o}_{Fe^{III}O}$ were required to reproduce the liquidus in the FeO–Fe_2O_3 region. It is mostly the balance between $\Delta g^{\circ}_{Fe^{II}O}$ and $\Delta g^{o}_{Fe^{III}O}$ that defines the shape of the liquidus and equilibrium $P(O_2)$ in this region. A smaller Bragg–Williams parameter $q^{001}_{Fe^{II},Fe^{III}(O)}$ was also required to fit the liquidus more accurately. The miscibility gap between metallic and oxide liquids was described by a positive parameter $g^{10}_{Fe^{II}O}$. Two positive parameters, $g^{20}_{Fe^{III}O}$ and $\Delta g^{o}_{Fe^{II}Fe^{III}}$, were set rather arbitrarily in order to suppress the formation of Fe^{III} in the Fe–FeO region. During the optimization, all available experimental points were considered simultaneously.

As can be seen from Figs. 1, 10, 11 and 13–16, the experimental data on the miscibility gap, phase boundaries and equilibrium oxygen partial pressures are well reproduced by the model. The calculated invariant points in the Fe–O system are given in Table 2.

4. Conclusions

A complete reevaluation of the Fe–O system has been performed. Contrary to the earlier assessments, it takes into account

Table 2
Calculated invariant points in the Fe–O system.

Reaction upon cooling	T, °C	Composition of the respective phases, at% O			$\log_{10}[P(O_2)$, atm]	Reaction type
L1 → Fe(bcc) + L2	1528	0.55	0.029	50.9	−8.75	Monotectic
Fe(bcc) → Fe(fcc) + L2	1391	0.021	0.0098	51.1	−9.84	Catatectic
L2 → Wüstite + Fe(fcc)	1371	51.1	51.2	0.0093	−10.02	Eutectic
Fe(fcc) + Wüstite → Fe(bcc)	912	0.0008	51.3	0.0014	−16.51	Peritectoid
Wüstite → Fe(bcc) + Magnetite	561	51.4	0.00002	57.1	−26.27	Eutectoid
L2 + Magnetite → Wüstite	1419	53.9	57.1	54.4	−6.05	Peritectic
L2 → Magnetite	1595		57.3		−1.06	Congruent melting
L2 → Magnetite + Gas (1 atm)	1582	58.2	57.6	100	0	Eutectic type
Magnetite + Gas (1 atm) → Fe_2O_3	1459	58.0	100	60.0	0	Peritectic type
L2 → Magnetite + Fe_2O_3	1552	58.9	58.1	60.0	0.88	Eutectic
L2 → Fe_2O_3	1620		60.0		5.11	Congruent melting

the measurements of the heat capacity and standard entropy of wüstite at several compositions [9,10], which substantially constrain the thermodynamic properties of this phase. In particular, these data made it possible to eliminate the uncertainty related to modeling of the configurational entropy of wüstite. The obtained good description of the experimental heat capacities of wüstite is important for calculating heat balances, while the phase diagrams and equilibrium oxygen partial pressures are reproduced equally well as in the earlier optimization of Sundman and Selleby [5,6]. This results in thermodynamic properties that are better constrained and more accurate than in the previous assessments not only for wüstite, but also for all other phases.

A model for the liquid phase over the whole composition range has been developed within the framework of the Quasichemical Formalism. This model describes simultaneously metallic and oxide liquids. It reflects the existence of two ranges of maximum short-range ordering at approximately FeO and Fe_2O_3 compositions.

Parameters of thermodynamic models for the liquid and solid phases have been optimized to provide the best description of all available thermodynamic and phase equilibrium data at a total pressure of 1 atm. The obtained self-consistent set of model parameters can be used to calculate the solubility of oxygen in liquid and solid iron. All experimental data are reproduced within experimental uncertainties.

Appendix A. Suplementary material

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.calphad.2014.12.005.

References

[1] H.A. Wriedt, J. Phase Equilib. 12 (1991) 170–200.
[2] R.P. Goel, H.H. Kellogg, J. Larrain, Metall. Trans. 11B (1980) 107–117.
[3] B. Bjorkman, CALPHAD 9 (1985) 271–282.
[4] M. Kowalski, P.J. Spencer, CALPHAD 19 (1995) 229–243.
[5] B. Sundman, J. Phase Equilib. 12 (1991) 127–140.
[6] M. Selleby, B. Sundman, CALPHAD 20 (1996) 381–392.
[7] P. Wu, G. Eriksson, A.D. Pelton, M. Blander, ISIJ Int. 33 (1993) 26–35.
[8] S.A. Decterov, E. Jak, P.C. Hayes, A.D. Pelton, Metall. Mater. Trans. 32B (2001) 643–657.
[9] F. Gronvold, S. Stolen, P. Tolmach, E.F.J. Westrum, J. Chem. Thermodyn. 25 (1993) 1089–1117.
[10] S. Stolen, R. Glockner, F. Gronvold, T. Atake, S. Izumisawa, Am. Miner.. 81 (1996) 973–981.
[11] L.S. Darken, R.W. Gurry, J. Am. Chem. Soc. 68 (1946) 798–816.
[12] P.A. Distin, S.G. Whiteway, C.R. Masson, Can. Metall. Q. 10 (1971) 13–18.
[13] W.A. Fischer, J.F. Schumacher, Arch. Eisenhuttenwes. 49 (1978) 431–435.
[14] C.R. Taylor, J. Chipman, Trans. AIME 154 (1943) 228–247.
[15] P. Vallet, C. Carel, Bull. Alloy. Phase Diagr. 10 (1989) 209–218.
[16] M. Hillert, B. Jansson, B. Sundman, Z. Metallkd. 79 (1988) 81–87.
[17] Y. Fei, S.K. Saxena, Phys. Chem. Miner. 13 (1986) 311–324.
[18] L. Kjellqvist, M. Selleby, J. Phase Equilib. Diff. 31 (2010) 113–134.
[19] L. Kjellqvist, M. Selleby, B. Sundman, CALPHAD 32 (2008) 577–592.
[20] A.V. Khvan, O. Fabrichnaya, G. Savinykh, R. Adam, H.J. Seifert, J. Phase Equilib. Diff. 32 (2011) 498–511.
[21] R. Luoma, Acta Polytech. Scand., Chem. Technol. Ser. 292 (2002) 1–91.
[22] A.D. Pelton, S.A. Decterov, G. Eriksson, C. Robelin, Y. Dessureault, Metall. Mater. Trans. 31B (2000) 651–659.
[23] A.D. Pelton, P. Chartrand, Metall. Mater. Trans. 32A (2001) 1355–1360.
[24] I.-H. Jung, S.A. Decterov, A.D. Pelton, Metall. Mater. Trans. 35B (2004) 493–507.
[25] S.A. Decterov, Y.-B. Kang, I.-H. Jung, J. Phase Equilib. Diffus. 30 (2009) 443–461.
[26] S.A. Decterov, E. Jak, I.-H. Jung, Y.-B. Kang, H. Kim, A.D. Pelton, Oxide database development for FactSage, CALPHAD XXX (2001) 82 (27 May-1 June 2001: York, UK).
[27] I.-H. Jung, S.A. Decterov, A.D. Pelton, Metall. Mater. Trans. 38B (2004) 877–889.
[28] E. Jak, P. Hayes, A.D. Pelton, S.A. Decterov, Int. J. Mater. Res. 98 (2007) 847–854.
[29] I.-H. Jung, S.A. Decterov, A.D. Pelton, Int. J. Mater. Res. 98 (2007) 816–825.
[30] C.W. Bale, E. Belisle, P. Chartrand, S.A. Decterov, G. Eriksson, K. Hack, I.-H. Jung, Y.-B. Kang, J. Melancon, A.D. Pelton, C. Robelin, S. Petersen, CALPHAD 33 (2009) 295–311.
[31] G.A. Waychunas, Rev. Miner. 25 (1991) 11–68.
[32] S.S. Todd, K.R. Bonnickson, J. Am. Chem. Soc. 73 (1951) 3894–3895.
[33] J.P. Coughlin, E.G. King, K.R. Bonnickson, J. Am. Chem. Soc. 73 (1951) 3891–3893.
[34] J. Rogez, J.F. Marucco, R. Castanet, J.C. Mathieu, Ann. Chim. (Paris) 7 (1982) 63–71.
[35] M.F. Marion, La Doc. Met. 24 (1955) 87–129.
[36] J. Ackermann, R.W. Sandford, Jr., A thermodynamic study of the wüstite phase, USAEC Rep. ANL 7250, Chemistry (TID-4500), Argonne, IL, USA, 1966, 46 pp.
[37] F.E. Rizzo, J.V. Smith, J. Phys. Chem. 72 (1968) 485–488.
[38] P. Vallet, P. Raccah, Rev. Metall., Mem. Sci. 62 (1965) 1–29.
[39] R.A. Giddings, R.S. Gordon, J. Am. Ceram. Soc. 56 (1973) 111–116.
[40] C. Picard, M. Dode, Bull. Soc. Chim. Fr. 7 (1970) 2486.
[41] A.A. Lykasov, Y.S. Kuznetsov, E.I. Pil'ko, V.J. Shishkov, V.A. Kozheurov, Russ. J. Phys. Chem. 43 (1969) 1754–1756.
[42] B. Swaroop, J.B. Wagner Jr., Trans. Metall. Soc. AIME 239 (1967) 1215–1218.
[43] I. Bransky, A.Z. Hed, J. Am. Ceram. Soc. 51 (1968) 231–232.
[44] R.L. Levin, J.B.J. Wagner, Trans. Metall. Soc. AIME 239 (1967) 1215–1218.
[45] L.S. Darken, R.W. Gurry, J. Am. Chem. Soc. 67 (1945) 1398–1412.
[46] H.G. Sockel, H. Schmalzried, Ber. Bunsenges. Phys. Chem. 72 (1968) 745–754.
[47] P. Gerdanian, M. Dode, J. Chim. Phys. Phys. – Chim. Biol. 62 (1965) 1010–1017.
[48] J.F. Marucco, P. Gerdanian, M. Dode, J. Chim. Phys. Phys. – Chim. Biol. 67 (1970) 906–913.
[49] D.C. Hilty, W. Crafts, J. Met. 194 (1952) 1307–1312.
[50] S. Ueda, K. Yamaguchi, Y. Takeda, Mater. Trans. 49 (2008) 572–578.
[51] Y. Takeda, Oxygen potential measurement of iron silicate slag–copper–matte system, in: Proceedings of the 5th International Conference Molten Slags, Fluxes Salts'97, Iron and Steel Society, Warrendale, PA, USA, 1997, pp. 735–743.
[52] L.B. Pfeil, J. Iron Steel Inst., London 11 (1929) 501–560.
[53] L.B. Pfeil, J. Iron Steel Inst., London 13 (1931) 237–257.
[54] K. Sanbongi, Tohoku University, vol. 1, 1949, pp. 213–222.
[55] N.A. Hovgard, P.N. Jensfelt, Jernkontor. Ann. 140 (1956) 467–493.
[56] C. Offroy, J. Aubry, Mem. Sci. Rev. Met. 60 (1963) 476–480.
[57] G.B. Barbi, J. Phys. Chem. 68 (1964) 2912–2916.
[58] R.A. Meussner, I.E. Richards, C.T. Fujii, Studies of high temperature materials (properties of FeO), Naval Research Laboratory Report, Washington, DC, 1965, pp. 26–28.
[59] C.T. Fujii, R.A. Meussner, Trans. Met. Soc. AIME 242 (1968) 1259–1265.
[60] H.F. Rizzo, R.S. Gordon, I.B. Cutler, J. Electrochem. Soc. 116 (1969) 266–274.
[61] J. Campserveux, G. Boureau, C. Picard, P. Gerdanian, Rev. Int. Hautes Temp. Refract. 6 (1969) 165.
[62] J. Campserveux, G. Boureau, C. Picard, P. Gerdanian, Ann. Chim. 5 (1970) 250–260.
[63] H. Asao, K. Ono, A. Yamaguchi, J. Moriyama, Mem. Fac. Eng., 66–77.
[64] J.F. Marucco, C. Picard, P. Gerdanian, M. Dode, J. Chim. Phys. Phys. – Chim. Biol.

67 (2014) 914–916.
[65] R.A. Giddings, R.S. Gordon, J. Electrochem. Soc. 121 (1974) 793–800.
[66] E. Takayama, N. Kimizuka, J. Electrochem. Soc. 127 (1980) 970–976.
[67] J.A. Barbero, M.A. Blesa, A.J.G.Z. Maroto, Phys. Chem. 124 (1981) 139–147.
[68] G. Chaudron, Ann. Chim. 16 (1921) 221.
[69] R. Schenck, T. Dingmann, Z. Anorg. Allg. Chem. 166 (1927) 113.
[70] J. Chipman, S. Marshall, J. Am. Chem. Soc. 62 (1940) 299–305.
[71] Y.D. Tretyakov, K.G. Khomyakov, Russ. J. Inorg. Chem. 7 (1962) 628–631.
[72] P. Vallet, P. Raccah, C.R. Hebd, Seances Acad. Sci. 258 (1964) 3679–3682.
[73] R.E. Johnson, A. Muan, Trans. Met. Soc. AIME 239 (1967) 1931–1937.
[74] G.G. Charette, S.N. Flengas, J. Electrochem. Soc.: Electrochem. Sci. 115 (1968) 796–804.
[75] K. Kodera, I. Kusunoki, M. Watanabe, Bull. Chem. Soc. Jpn. 42 (1969) 3036–3037.
[76] N. Tiberg, A. Muan, Metall. Trans. 1 (1970) 435–439.
[77] F. Abbattista, A. Burdese, M. Maja, Rev. Int. Hautes Temp. Refract. 12 (1975) 337–342.
[78] E. Jacobsson, E. Rosen, Scand. J. Metall. 10 (1981) 39–43.
[79] O. Sjoden, S. Seetharaman, L.I. Staffansson, Metall. Trans. 17 (1986) 179–184.
[80] H.S.C. O'Neill, Am. Miner. 73 (1988) 470–486.
[81] W.A. Fischer, G. Pateisky, Arch. Eisenhuttenwes. 41 (1970) 661–673.
[82] M. Fredriksson, E. Rosen, Scand. J. Metall. 13 (1984) 95–97.
[83] M.N. Dastur, J. Chipman, Trans. Am. Inst. Min., Metall. Pet. Eng. 1 (1949) 441–445.
[84] K. Schwerdtfeger, Trans. Met. Soc. AIME 239 (1967) 1276–1281.
[85] C. Gatellier, M. Olette, C.R. Acad, Sci., Paris, Ser. C 266 (1968) 1133–1135.
[86] E.S. Tankins, N.A. Gokcen, G.R. Belton, Trans. Am. Inst. Min., Metall. Pet. Eng. 230 (1964) 820–827.
[87] T.P. Floridis, J. Chipman, Trans. AIME 212 (1958) 549–553.
[88] N.A. Gokcen, J. Met. 206 (1956) 1558–1567.
[89] H. Sakao, K. Sano, Nippon Kinzoku Gakkaishi 23 (1959) 671–674.
[90] W.A. Fischer, D. Janke, W. Ackermann, Arch. Eisenhuttenwes. 41 (1970) 361–367.
[91] Steelmaking Data Sourcebook, 19th Committee on Steelmaking, Gordon & Breach Science, Japan Society for the Promotion of Science, New York, NY, 1988.
[92] D. Shishin, S.A. Decterov, CALPHAD 38 (2012) 59–70.
[93] M.T. Hepworth, R.P. Smith, E.T. Turkdogan, Trans. Metall. Soc. AIME 236 (1966) 1278–1283.
[94] J.H. Swisher, E.T. Turkdogan, Trans. Metall. Soc. AIME 239 (1967) 426–431.
[95] A.T. Dinsdale, CALPHAD 15 (1991) 317–425.
[96] I. Ansara, Å. Jansson, Assesment of the Copper–Iron System, Report Trita-Mac-0533, Materials Research Center, The Royal Institute of Technology, Stockholm, Sweden, 1993.
[97] A.D. Pelton, Thermodynamics and Phase Diagrams, in: Fifth Edition,in: D. E. Laughlin, K. Hono (Eds.), Physical Metallurgy, vol. 1, Elsevier B.V, 2014, pp. 203–303.